Youth Autistic Traits and Brain Activation States During Socioemotional Processing

Carolyn Lasch, Ph.D.,[1] Andrew T. Goldberg, B.A.,[1] Deanna M. Barch, Ph.D.,[1,2,3] Natasha Marrus, M.D., Ph.D.,[1,2] M. Catalina Camacho, Ph.D.,[1] Chad M. Sylvester, M.D., Ph.D.[1,3,4]

Objective: Different aspects of social-emotional processing are linked to unique whole-brain patterns of neural activity termed "activation states." Autistic traits in youths include impairments in socioemotional processing that vary depending on age and comorbidity. Characterizing differences in activation states associated with these impairments may explain why youths with high autistic traits process socioemotional information differently than peers.

Methods: This study examined the frequency of three previously identified activation states in 545 youths (ages 5–15 years, 39% female, 54% White, 68% non-Hispanic) from the Healthy Brain Network study as they watched a socially engaging movie. Regressions examined associations between autistic traits (measured by the Social Responsiveness Scale, 2nd ed.) and overall time spent in activation states, and whether these associations varied with age or co-occurring attention deficit hyperactivity disorder (ADHD) and anxiety symptoms (measured by the Strengths and Weaknesses Assessment of ADHD and Normal Behavior and the Screen for Child Anxiety Related Disorders).

Results: At younger ages (<9 years), children with more autistic traits spent more time in an activation state associated with increased activation in somatomotor and visual networks. At older ages (>14 years), children with more autistic traits spent more time in an activation state characterized by greater default mode and ventral attention network activation. Across all ages, children with more ADHD symptoms spent less time in an activation state associated with greater cingulo-opercular network activation.

Conclusions: Autistic traits are associated with differences in brain activation during socioemotional processing that vary with age and co-occurring mental health symptoms. Youths with more autistic traits may experience an altered developmental trajectory of social-emotional processing compared to peers with fewer autistic traits. Results inform personalized interventions and progress monitoring for youths with social impairments.

Am J Psychiatry 2026; XX:1–11; doi: 10.1176/appi.ajp.20251035

Autism spectrum disorder (ASD) includes impairments in socioemotional processing that vary with age and co-occurring symptoms. The brain bases of these impairments are incompletely understood. Brain activity during socioemotional processing can be characterized as entering into a limited set of "activation states," specific patterns of increases and decreases in activity across the brain (1, 2). Individuals vary in their tendency to enter specific brain activation states, likely reflecting individual differences in cognitive and emotional processing. Yet, it is not known how autistic traits (ATs; behaviors and personal qualities that are consistent with the phenotype of autism and are continuously distributed in the population even at subclinical levels) relate to individual differences in how much time is spent in different activation states or how these relations vary with age or comorbidity. Clarifying these relationships may help identify mechanisms of impaired socioemotional processing over different phases of development in children with ATs and provide new targets for treatments.

At a macroscale, whole-brain activity tends to fluctuate between distinct activation states during real-world socioemotional processing. Each activation state includes increases in regional brain activity and is thought to reflect a particular type of socioemotional or cognitive processing. For example, when viewing emotionally intense content, individuals tend to shift into an activation state that includes increased activity in the default mode network (3). When viewing scenes with more spoken dialogue, most individuals enter into an activation state that includes auditory and language networks (4). However, individuals can vary in when they enter into and transition between specific brain states. Furthermore, this variation aligns with individual differences in subjective experiences (2, 4) and with objective changes in arousal (4, 5). Examining patterns of activation states in individuals with elevated rates of psychopathology may therefore help uncover mechanisms of differences in socioemotional processing.

Children with ASD and elevated ATs differ in behaviors and brain activity linked to socioemotional processing compared to other children. ASD affects social communication and functioning, with early emerging (6, 7) and persistent differences in social visual attention and cognition, potentially influencing emergence of later symptoms and altering or limiting trajectories of social attention and functioning later in development (i.e., canalization) (8). Neural differences, such as deviation in resting-state functional connectivity, emerge early (9, 10) with connectivity differences during social tasks identified in childhood and adolescence (11). The relations between ASD and brain and behavior, however, depend on comorbid psychopathology (12, 13) and developmental context. For example, attention deficit hyperactivity disorder (ADHD) and anxiety co-occur in ASD in 26%–48% and 11%–40% of cases, respectively (14–17) and may influence social attention and its underlying neural processes.

Variation in brain activation states elicited in different contexts may explain differences in socioemotional processing linked to ATs. These processing differences may vary with age and co-occurring symptoms. In previous work (18), we examined brain activation states in children ages 5–15 years from the Healthy Brain Network (HBN) cohort as they watched the movie *Despicable Me*, and we found three unique activation states. We observed that children with high generalized anxiety symptoms were more likely than peers to enter into a state that included high activity in the ventral attention network and default mode network, an activation state we theorized to be involved in processing emotionally salient and negatively valenced stimuli. We proposed that increased time in this brain state may contribute to hypervigilance in children with high anxiety and help explain cascading biases in information processing and subjective experience. Clarifying the activation state patterns of children with elevated ATs might similarly help explain how and why socioemotional processing is impacted in children with elevated ATs and how this varies with age and co-occurring symptoms. Such information would provide clear new targets for interventions, as activation states could be modified in a context-specific manner through therapy, cognitive training, or noninvasive neurostimulation.

Our goals in this study were to investigate the relationship between ATs and the distribution of participants' time across three previously identified brain activation states during the viewing of a socially engaging movie. We built on our prior work described above, which examined brain activation states as children in the HBN watched a movie. We examined associations between ATs and overall time spent in activation states, as well as whether the relations between activation states and ATs varied with age. Given the relatively high rates of psychopathology (e.g., 54.8% with ADHD, 45.4% with anxiety, 13.8% with ASD [19]) and comorbidities in the sample, we also assessed variation by co-occurring ADHD and anxiety symptoms. The results clarify how children with autistic symptoms process socioemotional information over development and suggest new avenues for developing brain-based treatments.

METHODS

Participants

The study participants constitute a subset of youths who participated in the HBN study (20) in the New York City metro area. Data were collected between winter 2016 and winter 2020. All participants provided assent and caregivers provided consent. All procedures were approved by the relevant site institutional review boards.

Brain activation state data derived from high-quality functional MRI (fMRI) data as described in our previous report (18) were available for 620 youths from the HBN's first nine data releases (the initial imaging sample in Table 1). Final analyses included 545 participants ages 5–15 for whom data were available from the Social Responsiveness Scale, 2nd edition (SRS) parent-report form (the final imaging and behavioral sample in Table 1). Participants with SRS data were older on average than those without SRS data (10.5 years vs. 9.71 years; $p=0.025$). The two groups did not differ significantly in time spent in any brain activation state (p values >0.45) or in ADHD or anxiety symptoms (p values >0.61). Table 1 summarizes participants' demographic characteristics; see Table S2 in the online supplement for additional details regarding participant ethnicity, Table S3 for pubertal stage, and Table S4 for caregiver-reported medication use during fMRI collection.

Measures

Autistic traits were measured with total t score from the parent-report version of the SRS (21), which includes items measuring social communication, cognition, awareness, motivation, and restricted and repetitive behaviors and has been validated in children ages 4–18 years. ADHD symptoms were measured with the average parent-report score from the Strengths and Weaknesses Assessment of ADHD and Normal Behavior (SWAN) (22). Anxiety was measured with the total self-report score from the Screen for Child Anxiety Related Disorders (SCARED) (23), which was completed by children ages 7.7 and older (N=422). Scores ≥25 may indicate an anxiety diagnosis. Self-report was specifically used for anxiety symptoms given past work suggesting stronger associations between naturalistic anxiety behaviors and child self-report than parent-report (24) and possible caregiver underreport of children's internalizing symptoms specifically (e.g., reference 25, although see Tables S17–S19 in the online supplement for sensitivity analyses with parent-report of anxiety symptoms). All continuous measures were mean centered and z scored for statistical analyses.

Clinician consensus diagnostic information was available for 367 of the 545 participants. The diagnostic process in HBN (described in the HBN protocol paper [20]; additional documentation is available via the HBN data portal [19]) integrates information from available questionnaires, assessments, and diagnostic interview with a clinician. See Table S5 in the online supplement for participant diagnoses at study enrollment as determined by the HBN diagnostic team and parent-reported educational eligibility, and Table S6 for subscale scores for all measures.

TABLE 1. Demographic characteristics and brain activation state data of study participants[a]

Characteristic or measure	Initial imaging sample (N=620)		Final imaging and behavioral sample (N=545)	
	N	%	N	%
Female	251	40	210	38.5
Right-handed	465	75	409	75.1
Household income				
<$10,000	13	2	12	2.2
$10,000 to $19,999	13	2	13	2.4
$20,000 to $29,999	19	3	18	3.5
$30,000 to $39,999	28	5	25	4.6
$40,000 to $49,999	20	3	20	3.7
$50,000 to $59,999	11	2	10	1.8
$60,000 to $69,999	27	4	25	4.6
$70,000 to $79,999	13	2	12	2.2
$80,000 to $89,999	21	3	21	3.9
$90,000 to $99,999	22	4	21	3.9
$100,000 to $149,999	101	16	98	18.0
≥$150,000	182	29	182	33.4
Declined to specify or no income data available	150	24	88	16.1
Race				
Asian	15	2.4	15	2.8
Black or African American	67	10.8	64	10.3
Hispanic or Latino	50	8.1	47	8.6
Native Hawaiian or Other Pacific Islander	1	0.2	1	0.2
White	298	48.1	293	53.8
Other	9	1.5	9	1.7
Two or more races	105	16.9	102	18.7
Unknown or no race information available	75	12.1	14	2.6
Ethnicity				
Hispanic or Latino	137	22.1	132	24.2
Not Hispanic or Latino	382	61.6	373	68.4
Declined to specify	30	4.8	30	5.5
No ethnicity information available	71	11.5	10	1.8
	Mean	SD	Mean	SD
Age (years)	10.4	2.81	10.5	2.78
Psychopathology				
SRS parent-report total t score (N=545)			56.3	10.9
SWAN average score (Ns, 551 and 541)	0.37	0.96	0.37	0.96
SCARED self-report total score (N=422)			21.9	15.9
Percentage of retained frames	97.2	4.72	97.3	4.68
Mean frame displacement (of retained frames)	0.360	0.184	0.357	0.185
Percentage of time in state 1	38.5	6.68	38.5	6.63
Percentage of time in state 2	26.4	7.08	26.4	7.07
Percentage of time in state 3	32.4	3.99	32.4	3.95

[a] Participants reporting "Hispanic or Latino" race all reported "Hispanic or Latino" as ethnicity as well. "Hispanic or Latino" was included in both racial and ethnic counts to reflect how information was reported by participants at intake. Participant sex was dichotomously coded as "male" or "female" within Healthy Brain Network data. SCARED, Screen for Child Anxiety Related Disorders; SRS, Social Responsiveness Scale, 2nd ed., parent report; SWAN, Strengths and Weaknesses Assessment of ADHD and Normal Behavior.

Brain activation states. Brain activation states were derived from time series of fMRI data collected during in-scanner passive viewing of a movie via Hidden Markov modeling (HMM) approaches, as described in our previous report (18). fMRI data collection took place while participants watched a 10-minute clip from the movie *Despicable Me*, which includes scenes such as a tea party between Gru and the girls, minions engaging in hijinks, an argument between Gru and Dr. Nefario, and Gru preparing to steal the moon.

Analyses used data from two HBN MRI sites, CitiGroup Cornell Brain Imaging Center and the Rutgers University Brain Imaging Center. Only participants with at least 80% low-motion data (frames <0.9 FD) were retained in analyses, a standard threshold for task-based fMRI analyses, which has been demonstrated to balance maximization of power and minimization of noise from motion (26).

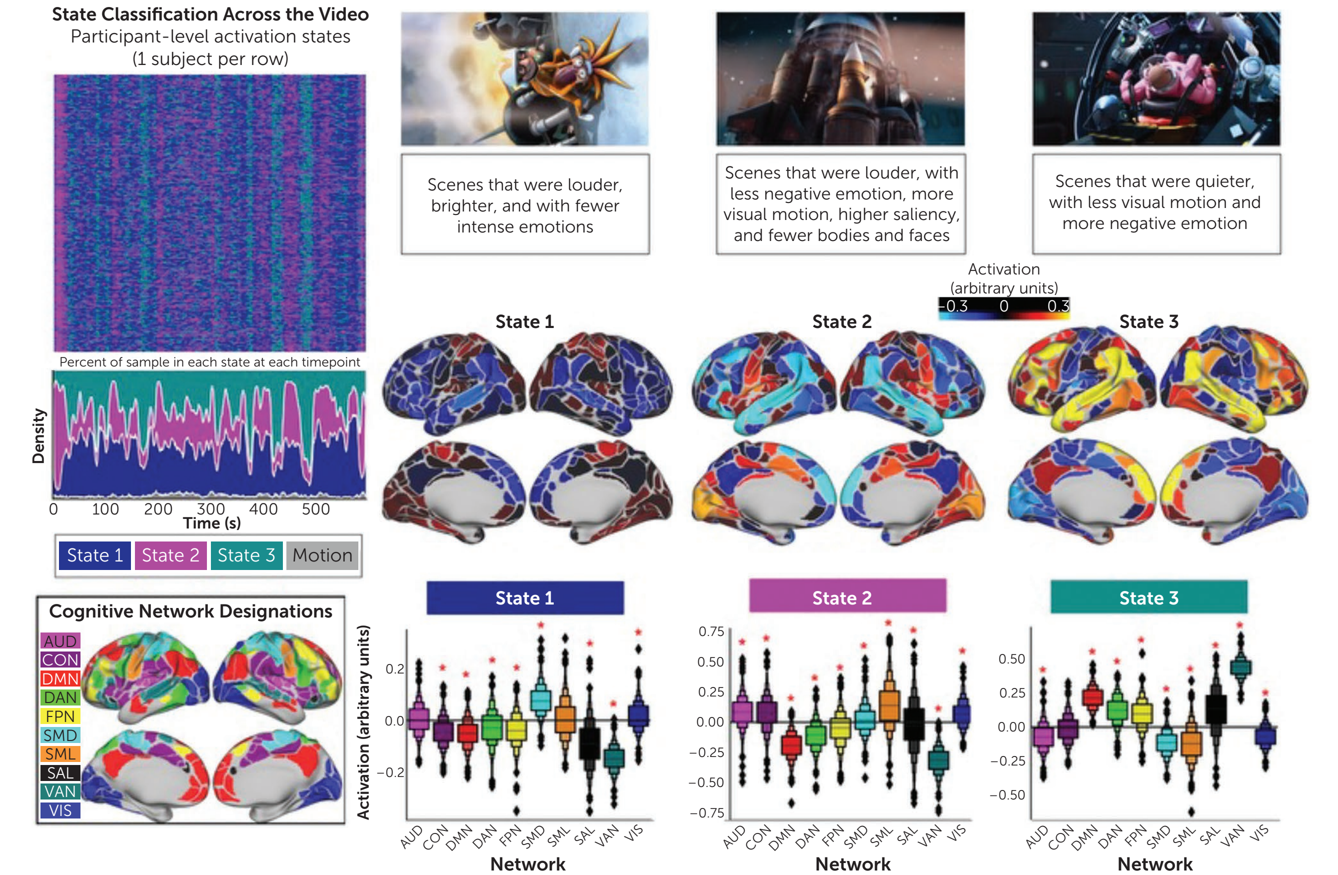
As described previously (18), HMM approaches suggested that a three-activation state model fit the data best based on the Akaike information criterion (AIC), the Bayesian information criterion, and permutation-based p values. States were characterized by relatively higher activation in specific networks. Briefly, state 1 was associated with higher activation in somatomotor and visual networks and lower activation in the cingulo-opercular network (CON), default mode network (DMN), dorsal attention network (DAN), frontoparietal network (FPN), ventral attention network (VAN), and salience network. State 2 was associated with higher activation in the auditory network, CON, somatomotor network, and visual network and lower activation in the DMN, DAN, FPN, salience network, and VAN. State 3 was associated with higher activation in the DMN, DAN, FPN, salience network, and VAN and lower activation in the auditory network, CON, somatomotor network, and visual network.

Activation states were assigned to each frame of each participant's time series based on this three-state model (see Figure S1 in the online supplement for examples of frame-level data and Figure S2 for comparison of state assignment posterior probabilities as a function of ATs), and total time in each state (excluding high-motion volumes) was calculated for each participant. See Figure S2 in the online supplement for age-related differences in percentage of time participants spent in each activation state. As described previously (18) and presented in Table S1 in the online supplement, variables such as scene loudness, brightness, and emotional content were associated with participants' likelihood of being in each activation state. Figure 1 includes representative examples of frames when most participants are in each activation state. Supplemental analyses indicated that ATs did not relate to participants' proportions of high-confidence frames of data (i.e., a probability <0.05 or >0.95 of being in a given state; p=0.54) (see Figure S3 in the online supplement).

Analytic Approach

For each of the three brain activation states, a similar series of analyses was performed. Five different models were calculated, representing five potential theories to explain

FIGURE 1. Network activation and movie features associated with three brain activation states[a]



[a] Upper left: brain state classification at each movie frame for each subject. Top middle: screen captures characteristic of each brain state. Bottom: brain state activation in cortical network space. AUD, auditory network; CON, cingulo-opercular network; DMN, default mode network; DAN, dorsal attention network; FPN, frontoparietal network; SMD, dorsal somatomotor network; SML, lateral somatomotor network; SAL, salience network; VAN, ventral attention network; VIS, visual network.

variation in the time spent in each activation state. They included:

1. A baseline model controlling for participant age, sex, and level of motion (as measured by mean frame displacement). This model is consistent with the theory that youth psychopathology and ATs are not associated with differences in activation state occupation.
2. A model with all covariates from model 1 as well as a main effect of SRS t score. This model is consistent with the theory that youths with higher ATs show consistent differences in state occupation across development.
3. A model with an SRS-by-age interaction term added to model 2. This model is consistent with the theory that the effects of ATs on state occupation vary across development.
4. A model with the main effect of SWAN scores added to model 3. This model is consistent with the theory that ADHD symptoms help explain variation in state occupation (in addition to developmental effects of ATs).
5. A model with the main effect of SCARED scores added to model 3. This model is consistent with the theory that anxiety symptoms help explain variation in state occupation (in addition to developmental effects of ATs).

We aimed to identify the model with the best fit and most plausibility using the corrected AIC [AICc] [27], which is better for quantifying model fits with smaller samples) and R^2. Differences in AICc (ΔAICc) of 2 or more indicate a significant difference in the plausibility of models (28). However, AICc can only compare fit of models across the same sample and sample size, and including variables of ADHD and anxiety symptoms in models 4 and 5 reduced the size of the available sample (to N=541 and N=422, respectively).

To avoid this reduction in sample size where possible, we elected to use p values of our effects of interest (SRS, SRS-by-age, SWAN, and SCARED) in the five initial models to identify a subset of “significant models” for each of the three activation states (see Tables S8, S10, and S13 in the online supplement). For each state, models identified as “significant” were then fitted in the same sample (reducing the sample size if models 4 or 5 were chosen), and AICc values were then used to determine which “significant model” most plausibly explained overall time in each activation state (see Tables S9, S11, and S14 in the online supplement). AICc weights indicate the probability that a given model is the best explanation for the given data. False discovery rate (FDR) correction was not performed in the initial assessment of model significance pre-AICc comparison, but see Tables S8, S10, and S13 in the online supplement for examination of significance after Benjamini-Hochberg (29) FDR correction.

Where appropriate, Johnson-Neyman intervals, which determine the moderator value at which a predictor becomes significant (30), were calculated for any relevant SRS-by-age interactions using the *interactions* R package (31). To keep models as parsimonious as possible, data collection site was not included as a predictor variable. Sensitivity analyses suggested that its inclusion in final models did not significantly alter effect sizes of variables of interest (i.e., age, ATs, and ADHD and anxiety symptoms; see Figure S4 in the online supplement).

To complement our main analyses, we computed exploratory post hoc frame-level regressions to assess associations between ATs and each participant's likelihood of being in a given state during specific moments of the clip (see Figures S5–S7 in the online supplement). Supplementary analyses also examined the effect of categorical ASD diagnosis in a subsample of participants for whom full diagnostic outcome information from HBN clinicians was available (N=357; see Figures S8–S13 in the online supplement).

RESULTS

Associations Among Autistic Traits, ADHD Symptoms, Anxiety Symptoms, and Age

Autistic traits measured with the SRS were linked to increased ADHD symptoms (SWAN; N=543; r=0.52, p<0.001) and increased anxiety symptoms (SCARED; N=420; r=0.14, p=0.003), but ATs were not linked to participant age (p=0.23) or sex (p=0.33). ADHD symptoms were lower in children who were older (N=539; r=−0.102, p=0.017) and higher in males (N=539; r=0.201, p<0.001). Anxiety symptoms were lower in males (N=420; r=0.155, p=0.001). ADHD and anxiety symptoms were not correlated (p=0.60). See Table S7 in the online supplement for pairwise correlations between symptoms and demographic variables, and diagnostic rates in sample pre-enrollment and poststudy diagnostic process. See Figure S2 in the online supplement for a visualization of age-related differences in percentage of time in each state as a function of participant ATs.

Associations Between Autistic Traits and Total Time in Each Brain State

State 1. State 1 is associated with higher activation in somatomotor and visual networks and is more likely to be elicited during scenes that are brighter, louder, and have fewer positive emotions. In comparing among models, the model with an SRS-by-age interaction (model 3) was marginally more plausible than the baseline model (AICc weights of 0.588 vs. 0.412, respectively) and explained more variance in R^2 (0.014 vs. 0.006; see Table S9 in the online supplement). In this model, children with more ATs spent more time in state 1 at younger ages, as indicated by a significant SRS-by-age interaction (β=−0.085, p=0.049). For children younger than 8.8 years old, ATs were significantly associated with more time in state 1 (Figure 2; see also Table S8 in the online supplement). There was no main effect of ATs on time spent in state 1, indicating that relations between ATs and state 1 varied with development.

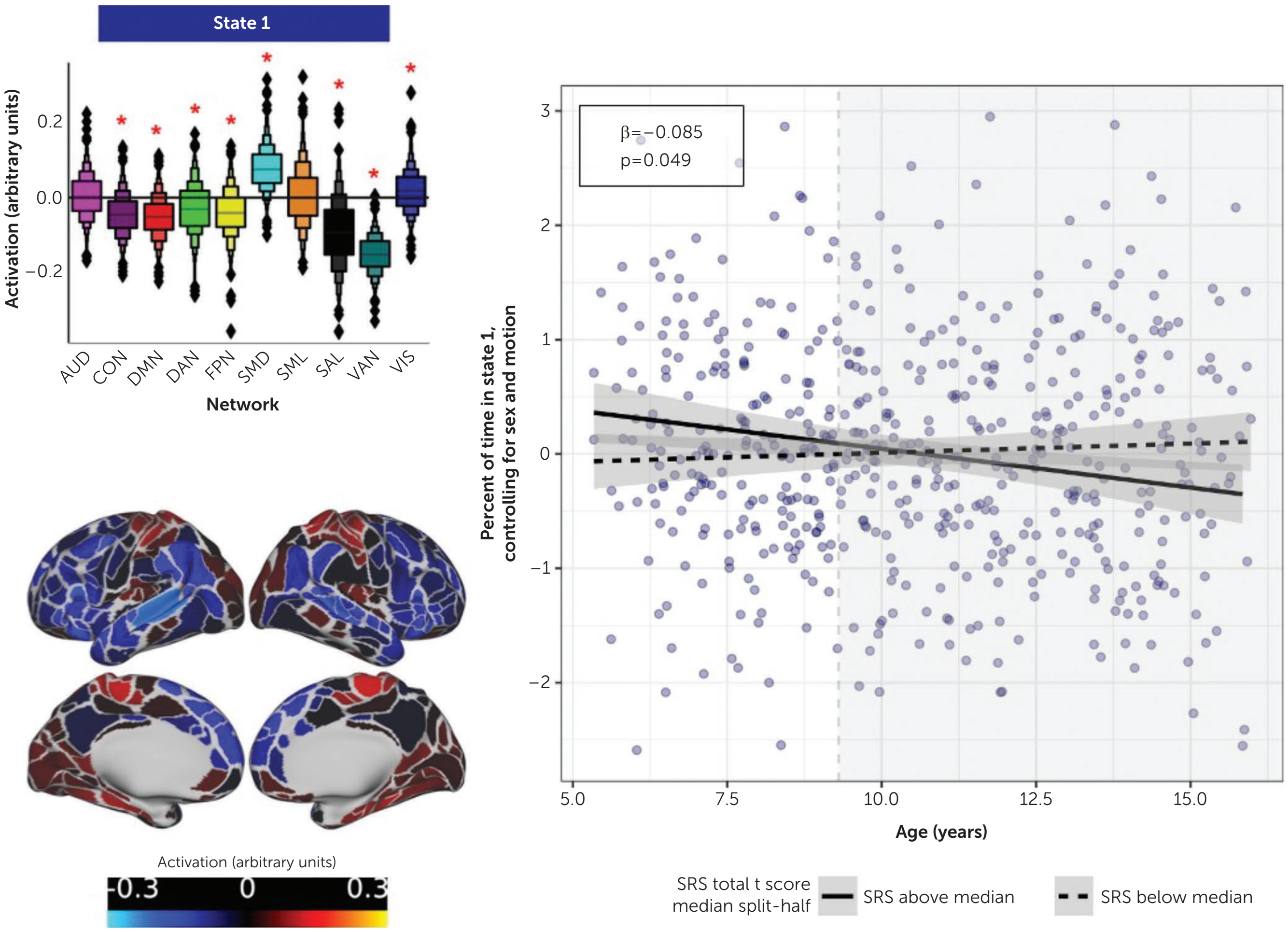
Neither ADHD nor anxiety helped explain variation in total time spent in state 1 (p values >0.18, models 4 and 5). Supplementary post hoc frame-level analyses suggested that scenes with significant age-by-SRS effects often included highly complex emotions and facial expressions or included high visual salience, such as bright colors and glowing lights (see Figure S5 in the online supplement).

State 2. State 2 is associated with higher activation in the auditory network, CON, somatomotor network, and visual network and is elicited during parts of the movie with less negative emotion and more motion, louder audio volume, and higher visual saliency. The most plausible model included the main effect of ADHD (model 4, AICc weight=0.746). It also explained the most variation based on R^2 values (0.099; see Table S11 in the online supplement). ADHD symptoms helped explain variance in time spent in state 2 (β=−0.104, p=0.036), such that participants with more ADHD symptoms spent less time in state 2 overall (Figure 3; see also Table S10 in the online supplement). In supplemental analyses using diagnostic categories rather than dimensional symptom measures, ADHD diagnosis showed an effect on state 2 occupation similar to that of ADHD symptoms (SWAN average score), such that participants with ADHD spent less time in state 2 than those without ADHD (see Figures S10 and S11 in the online supplement).

ATs were not associated with percentage of time spent in state 2 (p=0.051) in model 2 (see Table S10 in the online supplement). The SRS-by-age interaction was also not significant (p=0.83) (model 3), although the main effect of age was significant in all models (β values, 0.142–0.189, p values <0.001), such that older participants spent more time in state 2 than younger participants. Supplementary post hoc frame-level analyses suggested that in some scenes with high visual salience (where younger participants with higher ATs were more likely to be in state 1), older youths with higher ATs were more likely to be in state 2 (see Figure S6 in the online supplement).

State 3. State 3 is associated with higher activation in DMN, DAN, FPN, salience network, and VAN and was more likely to be elicited during parts of the video with lower audio volume, more negative emotions, and less motion. Of the preplanned analyses, the model with an SRS-by-age interaction provided the most plausible model for the data (model 3, AICc weight=0.473) and explained the most variation based on R^2

FIGURE 2. Activation pattern of state 1 and association between autistic traits, participant age, and total time spent in state[a]



[a] More autistic traits are associated with more total time in state 1 in younger participants (β=−0.085, p=0.049). Participants younger than 8.8 years old (non-gray region of the plot) with higher SRS scores (more autistic traits; solid line) spend more time in state 1 (greater activation in the somatomotor and visual networks) than their peers with lower SRS scores (dashed line). AUD, auditory network; CON, cingulo-opercular network; DMN, default mode network; DAN, dorsal attention network; FPN, frontoparietal network; SMD, dorsal somatomotor network; SML, lateral somatomotor network; SAL, salience network; SRS, Social Responsiveness Scale, 2nd ed.; VAN, ventral attention network; VIS, visual network.

values (0.052; see Table S14 in the online supplement). In this model, children with more ATs spent more time in state 3 at older ages, as indicated by a significant SRS-by-age interaction (β=0.087, p=0.041; see Table S13 in the online supplement). Participants age 14.3 years and older with more ATs spent more time in state 3 than their peers with fewer ATs (Figure 4). The main effect of ATs was not significant (p values >0.54).

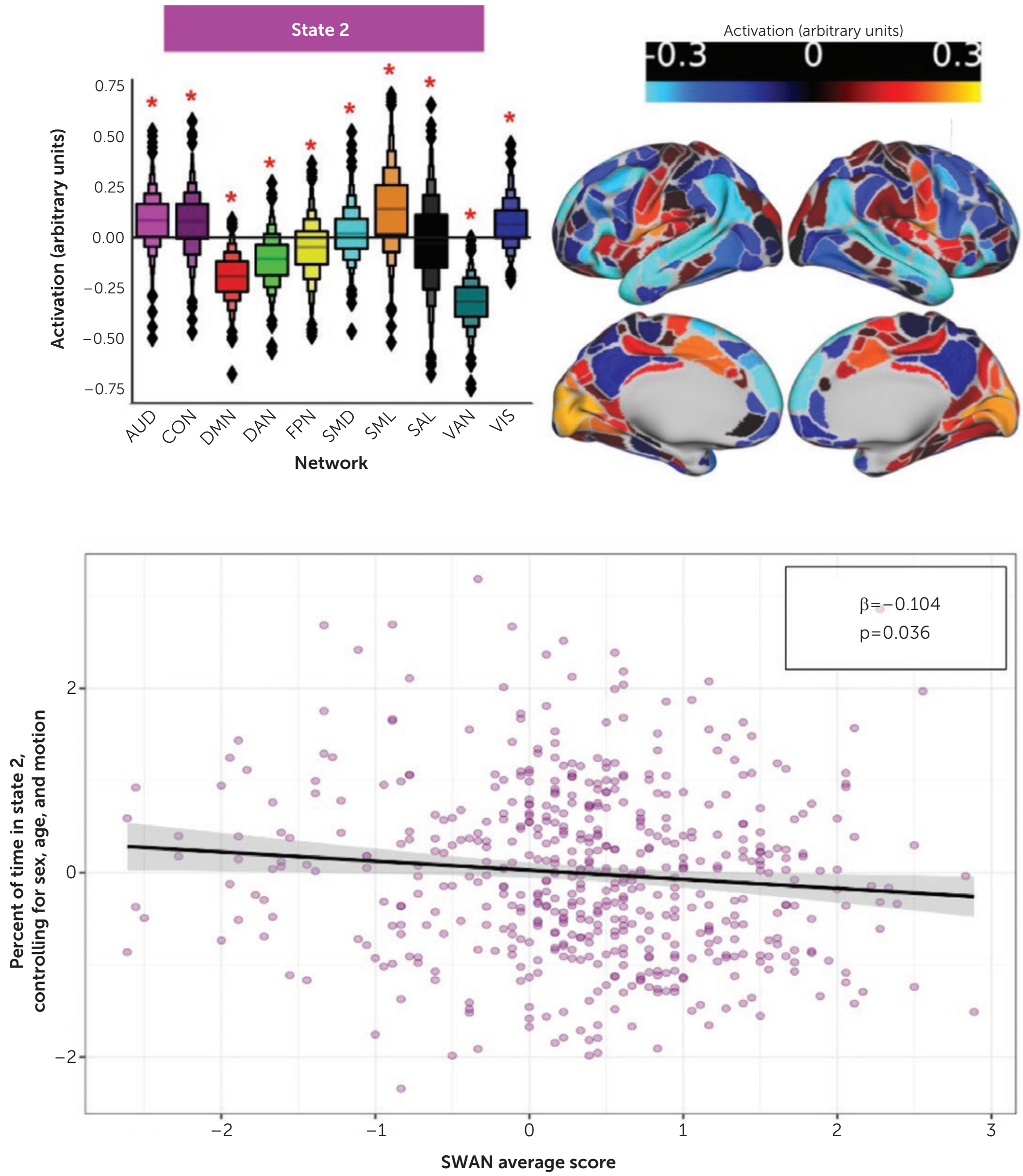
The main effects of anxiety and ADHD symptoms did not help explain the percentage of time spent in state 3 (p values >0.33), although supplemental analyses confirmed that significant effects of generalized anxiety identified in our previous report (18) persisted in a model with the SRS-by-age interaction and suggested a most plausible model with both significant main effects (see Tables S15 and S16 in the online supplement). Supplementary post hoc frame-level analyses suggested that age-by-AT effects were most significant during emotionally complex or intense scenes (see Figure S7 in the online supplement).

DISCUSSION

Summary of Findings

The results reveal that ATs are associated with age- and comorbidity-dependent differences in brain activation states linked to socioemotional processing. At younger ages (<9 years), children with more ATs spend more time in an activation state with increased activation in the somatomotor and visual networks, which may be associated with somatomotor processing and integration (state 1). Across all ages, children with more ADHD symptoms (commonly comorbid with ATs) spend less time in a state with increased activation in the CON, which may be associated with attention shifting and maintenance (state 2). At older ages (>14 years), children with more

FIGURE 3. Activation pattern of state 2 and association between ADHD symptoms and total time spent in state 2[a]



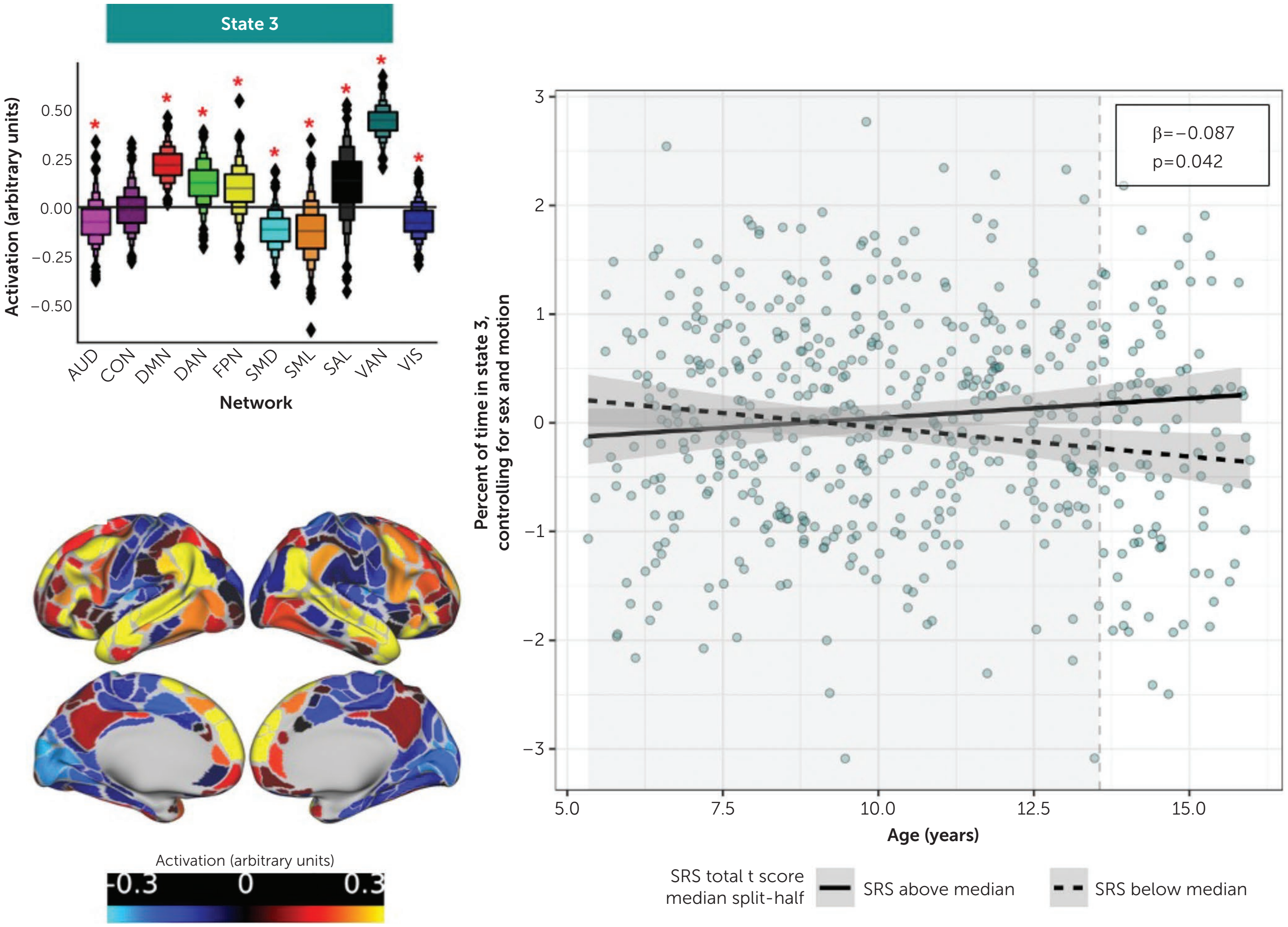
[a] A higher SWAN average score is associated with a smaller percentage of time in state 2 (β=−0.104, p=0.036). Participants with more ADHD symptoms spend less time in state 2 (greater activation in the cingulo-opercular, auditory, somatomotor, and visual networks). ADHD, attention deficit hyperactivity disorder; AUD, auditory network; CON, cingulo-opercular network; DMN, default mode network; DAN, dorsal attention network; FPN, frontoparietal network; SMD, dorsal somatomotor network; SML, lateral somatomotor network; SAL, salience network; SWAN, Strengths and Weaknesses Assessment of ADHD and Normal Behavior; VAN, ventral attention network; VIS, visual network.

ATs spend more time in an activation state with increased activation in the DAN and VAN, which may be associated with social processing and stimulus-driven attention (state 3).

Developmental cascades. This overall pattern of age-related results suggests that children high in ATs may follow a specific trajectory of socioemotional information processing over development that differs from that of children lower in ATs. The specific differences in socioemotional processing vary by age, suggesting that elevated ATs may be associated with an altered neurodevelopmental cascade that relates to age-specific challenges and enduring functional differences.

Before age 9, children with more ATs are more likely to be in a state that we theorize to be related to somatomotor processing and integration (state 1). This is consistent with previous work suggesting that even in infancy, autistic individuals attend to low-level salience in their visual environment over socially relevant content (6, 32), with differences persisting into childhood (33) and cascading across development, impacting social functioning (34).

FIGURE 4. Activation pattern of state 3 and association between autistic traits, participant age, and total time spent in state 3[a]



[a] Higher SRS total t score (more autistic traits) is associated with more time in state 3 (greater activation in the default mode, dorsal attention, frontoparietal, salience, and ventral attention networks), but only in older youths (β=−0.087, p=0.042). Youths 14.3 years old and older (non-gray region of plot) with more autistic traits (solid line) spend more time in state 3 than their peers with fewer autistic traits (dashed line). AUD, auditory network; CON, cingulo-opercular network; DMN, default mode network; DAN, dorsal attention network; FPN, frontoparietal network; SMD, dorsal somatomotor network; SML, lateral somatomotor network; SAL, salience network; SRS, Social Responsiveness Scale, 2nd ed.; VAN, ventral attention network; VIS, visual network.

Across all ages in the present study, children with more ADHD symptoms were less likely to be in a state that we theorize may be related to maintenance of focus and arousal (state 2). Given the high rates of ADHD in children with elevated ATs, many children with autism would thus experience reduced time in state 2 relative to peers. ADHD symptoms predicting decreased time in this state is consistent with the high rates of autistic children presenting with inattention, hyperactivity, and executive function struggles (35).

At age 14 and older, children with more ATs spend more time in an activation state that we theorize may be related to social processing and stimulus-driven attention (state 3). This suggests that in the teenage years, "decoding" social interactions maybe more effortful for youths with elevated ATs than for their peers. The more exaggerated social cues in animated interactions may allow older youths with more ATs to more readily engage with social-emotional content that is less subtle—but potentially less engaging for their peers with fewer ATs. These results are also consistent with previous research identifying different connectivity and activation patterns in autistic adolescents specifically during social processing tasks (11).

Overall, the present results build on previous findings and describe a cascade-like developmental trajectory of differences in socioemotional processing linked to ATs. Interestingly, the age-dependent effects in these results mirror varied presentations and complaints related to ASD across childhood and adolescence (36), providing additional face validity. Early concerns about ASD often arise based on differences in eye contact, joint attention, and repetitive gross motor movements such as rocking and flapping (37) (potentially related to the visual and somatomotor networks that are prominent in state 1). Similarly, autistic and neurodiverse adolescents are likely to report struggles with processing and responding to complex social cues (associated with the DMN and state 3) as a major struggle motivating an ASD assessment and diagnosis (38). Across childhood and adolescence, autistic individuals often present with concerns about attention, hyperactivity, and executive function, highlighting how both ASD and ADHD symptoms can impact an individual's attention, especially in a social context (related to the CON, prevalent in state 2). It will be important to replicate the present results in longitudinal samples to validate the finding that age-related differences are also present within individuals sampled across development.

Consideration of common ASD comorbidities and overlapping biological etiology. More broadly, examination of co-occurring symptoms is not only important in a sample like the HBN, with elevated prevalence rates for most diagnoses, but for understanding the effects of ASD generally. Diagnoses commonly co-presenting with ASD, such as ADHD and anxiety, further exacerbate qualitative differences in how autistic youths perceive, process, and experience social contexts (39) and worsen social and functional outcomes (40). ADHD, ASD, and anxiety co-occur frequently and are all implicated in early and persistent differences in neurodevelopment in overlapping functional brain networks, such as the DMN (41–43), suggesting shared neurobiological etiology. Taking ASD out of the context of these other symptoms may lead to a fundamental misunderstanding of the biological and neural mechanisms underlying its etiology and presentation, as exemplified in the findings of ADHD symptoms predicting time spent in state 2. Similarly, longitudinal research may be important in identifying separable effects of ATs, anxiety, and ADHD on network activation as well as where their overlapping symptomatology creates disorder-nonspecific impacts on processing and resulting functioning.

Developmentally Tailoring Interventions

While the presented analyses utilize cross-sectional data, they identify specific variations and shifts in developmental trajectories of brain activation associated with elevated ATs, which can inform more individually tailored and developmentally appropriate interventions. Younger children may benefit more from interventions that increase motivation and interest in the social aspects of the environment (i.e., decreasing time in a state related to sensory aspects including somatomotor and visual—state 1), thereby improving opportunities to gather important social information to aid in social cognition and functioning. Older youths may benefit more from interventions that support "decoding" layers of social information and meaning, thereby building skills to support more complex social interactions and functioning, which may potentially decrease time spent in a state with higher activation of the VAN and DMN—state 3. Youths of all ages can benefit from interventions (whether behavioral, pharmacological, or other) that improve the maintenance of attention to the social content of their environment (i.e., increasing time in state 2, which is associated with increased activation of the CON). Activation states could also be used to personalize and monitor interventions, such as identifying whether an individual tends to be in a less optimal activation state during specific aspects of social engagement and interaction (e.g., maintaining attention, "deciphering" emotional expressions), or to identify "phenotypes" of activation that cut across *DSM* diagnostic categories (44) and provide more straightforward treatment targets.

Consistency in Identified Brain States Across Samples and Methods

Others have identified similar activation states in fMRI data outside of HBN. Using adult task-based fMRI data from the Human Connectome Project, a recent study by Ye et al. (45) derived four unique states: low cognition (which resembles our state 1), fixation (which resembles our state 3), and two states, high cognition and cue/transition, which resemble our state 2. The study found good fit for these four states in HBN data as well as in another cross-sectional dataset of youths ages 8–21 with various clinical and cognitive profiles (46). States also fit well in longitudinal fMRI datasets (47), including when enriched for risk for substance use, and extending into early adulthood (48). Overall, this work supports the idea that brain states are replicable across datasets and different developmental periods (age 5–29). Ye et al. (45) also identified associations between variability in state occupation and executive function, suggesting that other metrics relating to brain activation states can be derived and linked to clinical and cognitive outcomes in future work. Both we and Ye and colleagues found good fit for our activation states in samples that include diverse phenotypes and clinical profiles. However, as research utilizing brain activation states progresses, it will be important to balance group-level characterization and interindividual variability in brain activation patterns and how they may relate to socioemotional experiences or clinically relevant symptoms. It will also be important to validate any interpretation of a given state (e.g., "concentration" or "low cognition") with objective measures.

Limitations and Future Directions

The Hidden Markov modeling–derived activation states used in our analyses were derived from a larger sample of pediatric data, representing a group average of participants' brain activation. Even if a child with elevated ATs is in the same state as a child with fewer ATs concurrently, we cannot know what specific aspects of the scene they viewed (e.g., visual saliency, emotional content) influenced each child's patterns of network coactivation. While frame-level analyses can identify specific scenes in which youths with more ATs were more or less likely to be in a given activation state (see Figures S5–S7 in the online supplement), we still cannot know that all youths with higher ATs were attending to the same aspects of a given scene to elicit that variation in activation states. Future work using in-scanner eye tracking can improve our knowledge of what emotional or social aspects of a given scene relate to variation in activation state occupation and whether this may change with development. Additionally, while these analyses investigated a wide age range, the cross-sectional nature of the data limits our ability to falsify hypotheses related to potential developmental cascades associated with the AT-by-age interactions we identified. Longitudinal data collection and analyses will be needed to confirm these theories.

Similarly, our analyses can only speak to how youths perceive a socially and emotionally engaged movie viewed while in an MRI scanner. Animated movies have many advantages compared to static face stimuli but are still removed from the demands of processing in vivo social stimuli embedded in the rich sensory and emotional context of the real world. Animated movies often utilize dramatic facial expressions and other social cues to clearly convey social meaning to viewers. Additionally, other content, such as music and sound effects (e.g., high-pitched strings to build tension) or lighting and coloring choices (e.g., a scene darkens to convey a character's negative emotion), further broadcast the emotions that filmmakers aim to evoke for the viewer. In-scanner movie viewing allows for viewing emotional and social content in the context of a movie and its plot and as such represents a marked improvement, in many ways, compared with static stimuli. However, it still differs in meaningful ways from real-world, first-person experiences. Future work should endeavor to assess socioemotional processing and its associations with ATs and other pediatric psychopathology in more naturalistic contexts.

While the present analyses focused on dimensional ATs, supplemental analyses using categorical diagnoses in a subsample of participants (N=357) did not find similar effects of ASD on activation states (see Figures S8–S13 in the online supplement). Investigation of categorical ASD diagnosis in this sample is also limited by the fact that HBN recruitment excluded nonverbal children and children with IQ <66 (who are disproportionately more likely to have an ASD diagnosis [20]) and children with low-functioning ASD. Future work can aim to replicate the activation states identified in our previous study (18) in a sample specifically recruited for meeting ASD diagnostic criteria. Additionally, while the present analyses were not preregistered, preregistration of future analyses can strengthen reproducibility and support transparency more broadly in neuroimaging and developmental neuroscience research.

This study adds to our understanding of the brain activation underlying socioemotional processing in youths with elevated ATs, within the context of development and common co-occurring symptoms. ATs influence total time spent in all three brain activation states assessed, with age-dependent effects related to states associated with visual and somatomotor networks in younger participants (state 1) and with the DMN and VAN in older participants (state 3). Our analyses also highlight how accounting for co-occurring symptoms such as those of ADHD and anxiety provide important context for the effects of ATs, such as decreased time in a state associated with the CON and attention and arousal (state 2). Collectively, findings related to all three states illustrate the possible cascade-like impacts of differences in brain function and connectivity on social processing and functioning for individuals with elevated ATs across childhood and adolescence. These findings highlight the utility of brain activation states in identifying and personalizing intervention targets as well as in monitoring treatment progress in youths with impairments in social functioning.

AUTHOR AND ARTICLE INFORMATION

[1]Department of Psychiatry, Washington University in St. Louis School of Medicine, St. Louis. [2]Department of Psychological and Brain Sciences, Washington University in St. Louis, St. Louis. [3]Department of Radiology, Washington University in St. Louis School of Medicine, St. Louis. [4]Taylor Family Institute for Innovative Psychiatric Research, Washington University in St. Louis School of Medicine, St. Louis.

Send correspondence to Dr. Lasch (lasch@wustl.edu).

Dr. Lasch https://orcid.org/0000-0002-3367-4445

Presented in part as a poster at the 13th Annual Flux Congress, Dublin, September 4–6, 2025.

The authors were supported by NIMH grants T32MH1000019, T32 GM007200, U01DA041120, R01 MH113883, U01 DA055367, F32 MH132185, R01MH131584, R01MH122389, and R01MH134966.

Dr. Barch has served as consultant for Johnson & Johnson Innovative Medicine and Boehringer Ingelheim. The other authors report no financial relationships with commercial interests.

Received September 29, 2025; revisions received March 18 and May 4, 2026; accepted May 11, 2026; published online August 12, 2026.

REFERENCES

1. Vemuri K, Surampudi BR: An exploratory investigation of functional network connectivity of empathy and default mode networks in a free-viewing task. Brain Connect 2015; 5:384–400
2. Song H, Finn ES, Rosenberg MD: Neural signatures of attentional engagement during narratives and its consequences for event memory. Proc Natl Acad Sci U S A 2021; 118:e2021905118
3. Chang LJ, Jolly E, Cheong JH, et al: Endogenous variation in ventromedial prefrontal cortex state dynamics during naturalistic viewing reflects affective experience. Sci Adv 2021; 7:eabf7129
4. van der Meer JN, Breakspear M, Chang LJ, et al: Movie viewing elicits rich and reliable brain state dynamics. Nat Commun 2020; 11:5004
5. Rosenberg MD, Scheinost D, Greene AS, et al: Functional connectivity predicts changes in attention observed across minutes, days, and months. Proc Natl Acad Sci U S A 2020; 117:3797–3807
6. Jones W, Klin A: Attention to eyes is present but in decline in 2–6-month-old infants later diagnosed with autism. Nature 2013; 504:427–431
7. Constantino JN, Kennon-McGill S, Weichselbaum C, et al: Infant viewing of social scenes is under genetic control and is atypical in autism. Nature 2017; 547:340–344
8. Girault JB: The developing visual system: a building block on the path to autism. Dev Cogn Neurosci. 2025; 73:101547
9. Liu J, Okada NJ, Cummings KK, et al: Emerging atypicalities in functional connectivity of language-related networks in young infants at high familial risk for ASD. Dev Cogn Neurosci 2020; 45:100814
10. Chen B, Linke A, Olson L, et al: Greater functional connectivity between sensory networks is related to symptom severity in toddlers with autism spectrum disorder. J Child Psychol Psychiatry 2021; 62:160–170
11. Herringshaw AJ, Kumar SL, Rody KN, et al: Neural correlates of social perception in children with autism: local versus global preferences. Neuroscience 2018; 395:49–59

12. You X, Norr M, Murphy E, et al: Atypical modulation of distant functional connectivity by cognitive state in children with autism spectrum disorders. Front Hum Neurosci 2013; 7:482
13. Bartolotti J, Sweeney JA, Mosconi MW: Functional brain abnormalities associated with comorbid anxiety in autism spectrum disorder. Dev Psychopathol 2020; 32:1273–1286
14. Mutluer T, Aslan Genç H, Özcan Morey A, et al: Population-based psychiatric comorbidity in children and adolescents with autism spectrum disorder: a meta-analysis. Front Psychiatry 2022; 13:856208
15. Rong Y, Yang C-J, Jin Y, et al: Prevalence of attention-deficit/hyperactivity disorder in individuals with autism spectrum disorder: a meta-analysis. Res Autism Spectr Disord 2021; 83:101759
16. Thiele-Swift HN, Dorstyn D-S: Anxiety prevalence in youth with autism: a systematic review and meta-analysis of methodological and sample moderators. Rev J Autism Dev Disord 2026; 13:76–89
17. van Steensel FJA, Bögels SM, Perrin S: Anxiety disorders in children and adolescents with autistic spectrum disorders: a meta-analysis. Clin Child Fam Psychol Rev 2011; 14:302–317
18. Camacho MC, Schwarzlose RF, Perino MT, et al: Youth generalized anxiety and brain activation states during socioemotional processing. JAMA Psychiatry 2025; 82:264–273
19. Health Brain Network: Phenotypic Data. Child Mind Institute, 2024. https://fcon_1000.projects.nitrc.org/indi/cmi_healthy_brain_network/File/_pheno/HBN_DiagnosticProcess_v3.pdf
20. Alexander LM, Escalera J, Ai L, et al: An open resource for transdiagnostic research in pediatric mental health and learning disorders. Sci Data 2017; 4:170181
21. Constantino JN, Todd RD: Autistic traits in the general population: a twin study. Arch Gen Psychiatry 2003; 60:524–530
22. Swanson JM, Schuck S, Porter MM, et al: Categorical and dimensional definitions and evaluations of symptoms of ADHD: history of the SNAP and the SWAN rating scales. Int J Educ Psychol Assess 2012; 10:51–70
23. Birmaher B, Khetarpal S, Brent D, et al: The Screen for Child Anxiety Related Emotional Disorders (SCARED): scale construction and psychometric characteristics. J Am Acad Child Adolesc Psychiatry 1997; 36:545–553
24. Bowers ME, Reider LB, Morales S, et al: Differences in parent and child report on the Screen for Child Anxiety-Related Emotional Disorders (SCARED): implications for investigations of social anxiety in adolescents. J Abnorm Child Psychol 2020; 48:561–571
25. Martin SR, Zeltzer LK, Seidman LC, et al: Caregiver-child discrepancies in reports of child emotional symptoms in pediatric chronic pain. J Pediatr Psychol 2020; 45:359–369
26. Siegel JS, Power JD, Dubis JW, et al: Statistical improvements in functional magnetic resonance imaging analyses produced by censoring high-motion data points. Hum Brain Mapp 2014; 35:1981–1996
27. Hurvich CM, Tsai C-L: Regression and time series model selection in small samples. Biometrika 1989; 76:297–307
28. Burnham KP, Anderson DR, Huyvaert KP: AIC model selection and multimodel inference in behavioral ecology: some background, observations, and comparisons. Behav Ecol Sociobiol 2011; 65:23–35
29. Benjamini Y, Hochberg Y: On the adaptive control of the false discovery rate in multiple testing with independent statistics. J Educ Behav Stat 2000; 25:60–83
30. Johnson PO, Fay LC: The Johnson-Neyman technique, its theory and application. Psychometrika 1950; 15:349–367
31. Long JA: Interactions: Comprehensive, User-Friendly Toolkit for Probing Interactions, 2024. https://cran.r-project.org/package=interactions. doi: 10.32614/CRAN.package.interactions
32. Jones W, Klaiman C, Richardson S, et al: Development and replication of objective measurements of social visual engagement to aid in early diagnosis and assessment of autism. JAMA Netw Open 2023; 6:e2330145
33. Chita-Tegmark M: Social attention in ASD: a review and meta-analysis of eye-tracking studies. Res Dev Disabil 2016; 48:79–93
34. Girault JB, Nishino T, Talović M, et al: Functional connectivity between the visual and salience networks and autistic social features at school-age. J Neurodev Disord 2025; 17:23
35. Tureck K, Matson JL, Cervantes P, et al: Autism severity as a predictor of inattention and impulsivity in toddlers. Dev Neurorehabil 2015; 18:285–289
36. Baird G, Douglas HR, Murphy MS: Recognising and diagnosing autism in children and young people: summary of NICE guidance. BMJ 2011; 343:d6360
37. Neurodevelopmental disorders; in Diagnostic and Statistical Manual of Mental Disorders, 5th ed, text revision. Washington, DC, American Psychiatric Association Publishing, 2022. https://doi.org/10.1176/appi.books.9780890425787.x01_Neurodevelopmental_Disorders
38. Trew S: The experience of autism from the perspective of autistic adolescents. Disabil Soc 2025; 40:1801–1831
39. Wan B, Wang Z, Jung M, et al: Effects of the co-occurrence of anxiety and attention-deficit/hyperactivity disorder on intrinsic functional network centrality among children with autism spectrum disorder. Autism Res 2019; 12:1057–1068
40. Chiang H-L, Gau SS-F: Comorbid psychiatric conditions as mediators to predict later social adjustment in youths with autism spectrum disorder. J Child Psychol Psychiatry 2016; 57:103–111
41. Sonuga-Barke EJS, Castellanos FX: Spontaneous attentional fluctuations in impaired states and pathological conditions: a neurobiological hypothesis. Neurosci Biobehav Rev 2007; 31:977–986
42. Eggebrecht AT, Elison JT, Feczko E, et al: Joint attention and brain functional connectivity in infants and toddlers. Cereb Cortex 2017; 27:1709–1720
43. Sylvester CM, Corbetta M, Raichle ME, et al: Functional network dysfunction in anxiety and anxiety disorders. Trends Neurosci 2012; 35:527–535
44. Lenze EJ, Nicol GE, Barbour DL, et al: Precision clinical trials: a framework for getting to precision medicine for neurobehavioural disorders. J Psychiatry Neurosci 2021; 46:E97–E110
45. Ye J, Tejavibulya L, Dai W, et al: Variation in moment-to-moment brain state engagement follows a consistent trajectory during development. Neuron 2025; 113:3863–3875.e6
46. Satterthwaite TD, Connolly JJ, Ruparel K, et al: The Philadelphia Neurodevelopmental Cohort: a publicly available resource for the study of normal and abnormal brain development in youth. Neuroimage 2016; 124:1115–1119
47. Schumann G, Loth E, Banaschewski T, et al: The IMAGEN study: reinforcement-related behaviour in normal brain function and psychopathology. Mol Psychiatry 2010; 15:1128–1139
48. Cope LM, Hardee JE, Martz ME, et al: Developmental maturation of inhibitory control circuitry in a high-risk sample: a longitudinal fMRI study. Dev Cogn Neurosci 2020; 43:100781